NUCLEAR POWER REACTORS

Small Modular Reactors

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- There is strong interest in small and simpler units for generating electricity from nuclear power, and for process heat.
- Small Modular Reactors (SMRs) represent a broad suite of smaller-scale designs that seek to apply the principles of modularity, factory fabrication, and serial production to nuclear energy.
- SMRs offer additional flexibility in operation and wider deployment opportunities, allowing for nuclear to be used in more locations and for a greater range of applications.

What is an SMR?

Small modular reactors (SMRs) are defined as nuclear reactors generally 300 MWe equivalent or less, designed with modular technology using module factory fabrication, pursuing economies of series production and short construction times.

A few small reactors are already in operation and there are dozens of designs in development.

Size

Traditional nuclear reactors produce around 1000 MWe or more of electricity per unit. SMRs, by contrast, are defined by their modest

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electrical output – typically less than 300 MWe. Some definitions extend to medium-sized reactors of up to 600 MWe.

The smaller capacity of SMRs allows for deployment in settings where large plants may not be practical – such as remote communities, industrial clusters, or regions with small electricity grids.

Modularity

The "modular" in small modular reactor refers to both the design and construction approach.

Both large and small nuclear reactors increasingly make use of modular construction techniques – assembling major components in factories or specialized facilities before transporting them to site. This approach improves quality control, shortens on-site construction times, and can reduce overall costs.

However, SMRs take modularity several steps further:

- Factory fabrication: components or even entire reactor modules are designed to be built in factories under controlled conditions.
- Serial production: SMR designers plan for serial production to achieve economies of series, similar to those achieved in the aerospace industry.
- On-site assembly: factory-fabricated modules are shipped to the site for assembly potentially reducing construction times and costs.
- Scalability: many SMRs are designed for incremental deployment, reducing financial risk and providing flexible solutions for customers and end users.

Reactor Technology

SMRs encompass a range of reactor types, many of which are evolutions of existing designs rather than entirely new technologies. Most SMRs that are candidates for near-term deployment are based on proven water-cooled reactor technologies.

Four main technology options are being pursued: light water reactors, fast neutron reactors, graphite-moderated high temperature reactors and various kinds of molten salt reactors

(MSRs). The first has the lowest technological risk, but the second (FNR) can be smaller, simpler and with longer operation before refuelling. Some MSRs are fast-spectrum.

Most SMRs are smaller, simplified LWRs and use the same type of low-enriched uranium fuel with water as coolant. However, some are fast reactors cooled by liquid metals such as sodium or lead. There are also high temperature gas-cooled designs and molten salt reactors in development. Some designs use advanced fuels with higher (5-20%) levels of enrichment (High Assay, Low Enriched Uranium, HALEU) or mixed oxide (MOX) fuel which means they can recycle some materials usually considered waste.

Drivers for SMRs

In the 1950s and 1960s nuclear power plants used reactors that now would be considered small. At that time, many diverse reactor designs were developed, with some being successfully demonstrated. However, from the 1970s onwards utilities increasingly settled on light-water cooled reactors (LWRs) and asked technology vendors to make them larger and larger to achieve economies of scale. Hundreds of units were built and these large units remain the mainstay of global nuclear power.

The current push for SMRs represents rebalancing of historic economies of scale towards economies of series production, as well as a wish to use nuclear energy for more applications than utility scale electricity generation.

Large-scale reactors provide reliable, 24/7 electricity supply on a low cost-per-unit basis, ideal for most on-grid applications. High upfront capital costs are followed by exceptionally low and stable operational costs, meaning it generally makes sense to operate large reactors continuously. Across the range of SMRs in development, new technical features coupled with smaller scale offer potential benefits, particularly in certain deployment scenarios:

- Affordability: by virtue of their smaller size, SMRs have a significantly lower capital outlay per unit than large-scale equivalents. This reduces financial risk and allows for a wider range of investors and owners of SMRs.

- Modular design: some SMRs are designed to be deployed in modules, allowing capacity to be scaled over time to match

demand.

- Operational flexibility: modern large-scale reactors can load-follow but are generally operated 24/7. Some SMRs are designed specifically with operational flexibility in mind, with, for example, the ability to quickly switch between electricity and heat generation.

- Deployment flexibility: SMRs expand the range of locations and deployment scenarios for nuclear, extending the benefits of nuclear to more end users. SMRs can be deployed on grids that would be too small to accommodate large reactors, whilst smaller core sizes and the passive safety features of some SMRs offer the promise of smaller emergency planning zones, allowing for deployment nearer urban areas and on industrial sites.

SMR development is proceeding in Western countries with a lot of private investment, including small companies. The involvement of these new investors indicates a profound shift taking place from government-led and -funded nuclear R&D to that led by the private sector and people with strong entrepreneurial goals, often linked to a social purpose. That purpose is often deployment of affordable clean energy, without carbon dioxide emissions.

SMR designs and projects

There are a range of over 100 SMR designs at various stages of development. For more information on SMR designs, please visit our Small Modular Reactor Design Database.

There are a small number of SMRs already operating and under construction, as well as numerous projects at various stages of deployment. For more information on deployment of SMRs, please visit: Small Modular Reactor (SMR) Global Tracker.

Fuel

The current fleet of nuclear reactors runs primarily on uranium fuel enriched up to 5% uranium-235 (U-235). High-assay low-enriched uranium (HALEU) is defined as uranium enriched to greater than 5% and less than 20% of the U-235 isotope. Applications for

HALEU are today limited to research reactors and medical isotope production. However, HALEU will be needed for many advanced power reactor fuels, and more than half of the small modular reactor (SMR) designs in development.

HALEU is not yet widely available commercially. At present only Russia and China have the infrastructure to produce HALEU at scale. Centrus Energy, in the United States, began producing HALEU from a demonstration-scale cascade in October 2023.

HALEU can be produced with existing centrifuge technology but requires a specific nuclear fuel cycle infrastructure and the development of new or modified regulations and licensing regimes. Moreover, new or modified transport containers will be required for the movement of the large quantities of HALEU required for the deployment of SMRs and advanced reactors.

Establishing the supply chain to produce and deliver HALEU to customers will require significant capital investment. Governments will need to play a role initially until demand from the commercial market provides a sufficient signal to support private investment.

For more information see page on: High-Assay Low-Enriched Uranium (HALEU).

Licensing and legislative support for SMRs

Licensing is potentially a challenge for SMRs, as design certification, construction and operation licence costs are not necessarily less than for large reactors. Several developers have engaged with the Canadian Nuclear Safety Commission's (CNSC's) pre-licensing vendor design review process, which identifies fundamental barriers to licensing a new design in Canada and assures that a resolution path exists. The pre-licensing review is essentially a technical discussion, phase 1 of which involves about 5000 hours of staff time, considering the conceptual design and charged to the developer. Phase 2 is twice that, addressing system-level design.

United States

In January 2012 the DOE called for applications from industry to support the development of one or two US light-water reactor designs, allocating $452 million over five years through the SMR Licensing Technical Support (LTS) programme. Four applications were made, from Westinghouse, Babcock & Wilcox, Holtec, and

NuScale Power, the units ranging from 225 down to 45 MWe. The DOE announced its decision in November 2012 to support the B&W 180 MWe mPower design, to be developed with Bechtel and TVA. Through the five-year cost-share agreement, the DOE would invest up to half of the total project cost, with the project's industry partners at least matching this. The total would be negotiated between the DOE and B&W, and the DOE had paid $111 million by the end of 2014 before announcing that funds were cut off due to B&W shelving the project. However B&W is not required to repay any of the DOE money, and the project, capped at $15 million per year, is now under BWX Technologies. The company had spent more than $375 million on the mPower programme to February 2016.

In March 2012 the DOE signed agreements with three companies interested in constructing demonstration small reactors at its Savannah River site in South Carolina. The three companies and reactors are: Hyperion (now Gen4 Energy) with a 25 MWe fast reactor, Holtec with a 160 MWe PWR, and NuScale with its 45 MWe PWR (since increased to 60 MWe and then to 77 MWe – see below). The agreements concerned the provision of land but not finance. The DOE was in discussion with four further small reactor developers regarding similar arrangements, aiming to have in 10-15 years a suite of small reactors providing power for the DOE complex. (Over 1953-1991, Savannah River was where a number of production reactors for weapons plutonium and tritium were built and run.)

In March 2013 the DOE called for applications for second-round funding, and proposals were made by Westinghouse, Holtec, NuScale, General Atomics, and Hybrid Power Technologies, the last two being for EM2 and Hybrid SMR, not PWRs. Other (non-PWR) small reactor designs will have modest support through the Reactor Concepts RD&D programme. A late application "from left field" was from National Project Management Corporation (NPMC) which includes a cluster of regional partners in the state of New York, South Africa's PBMR company, and National Grid, the UK-based grid operator with 3.3 million customers in New York, Massachusetts and Rhode Island.*

* The project is for an HTR of 165 MWe, apparently the earlier direct-cycle version of the shelved PBMR, emphasising its 'deep burn' attributes in destroying actinides and achieving high burn-up at high temperatures. The PBMR design was a contender with Westinghouse backing for the US Next-Generation Nuclear Power (NGNP) project, which has stalled since about 2010.

In December 2013 the DOE announced that a further grant would be made to NuScale on a 50-50 cost-share basis, for up to $217 million over five years, to support design development and NRC certification and licensing of its initially 45 MWe small reactor design, subsequently increased to 60 MWe and then 77 MWe. In mid-2013 NuScale launched the Western Initiative for Nuclear (WIN) – a broad, multi-western state collaboration – to study the demonstration and deployment of multi-module NuScale SMR plants in the western USA. WIN includes Energy Northwest (ENW) in Washington and Utah Associated Municipal Power Systems (UAMPS). It is now called the Carbon-Free Power Project. A demonstration NuScale SMR built as part of Project WIN was projected to be operational by 2024, at the DOE's Idaho National Laboratory (INL), with UAMPS as the owner and ENW the operator. This would be followed by a full-scale (originally 12- but now six-module) plant there owned by UAMPS, run by Energy Northwest, and costing $5000/kW on an overnight basis, hence about $3.0 billion, with an expected levelized cost of electricity (LCOE) of $58/MWh from 2030.

In January 2014 Westinghouse announced that was suspending work on its small modular reactors in the light of inadequate prospects for multiple deployment. The company said that it could not justify the economics of its SMR without government subsidies, unless it could supply 30 to 50 of them. It was therefore delaying its plans, though small reactors remain on its agenda. In 2016 however, the company was much more positive about SMRs. See also UK Support subsection below. However, in March 2017 BWXT suspended work on the mPower design, after Bechtel withdrew from the project.

The Small Modular Reactor Research and Education Consortium (SmrREC) has been set up by Missouri University of Science and Technology to investigate the economics of deploying multiple SMRs in the country. SmrREC has constructed a comprehensive model of the business, manufacturing and supply chain needs for a new SMR-centric nuclear industry.

Early in 2016 developers and potential customers for SMRs set up the SMR Start consortium to advance the commercialization of SMR reactor designs. Members of the consortium include Bechtel, BWX Technologies, Dominion, Duke Energy, Energy Northwest, Fluor, GE Hitachi Nuclear Energy, Holtec, NuScale, Ontario Power, PSEG Nuclear, Southern Nuclear, Tennessee Valley Authority (TVA) and UAMPS. The organization will represent the companies in interactions with the US Nuclear Regulatory Commission (NRC), Congress and the executive branch on small reactor issues. US

industry body the Nuclear Energy Institute (NEI) is assisting in the formation of the consortium, and is to work closely with the organization on policies and priorities relating to small reactor technology.

SMR Start has called for the DOE's LTS programme for SMRs to be extended to 2025 with an increase in funding. It pointed out: "Private companies and DOE have invested over $1 billion in the development of SMRs. However, more investment, through public-private partnerships is needed in order to assure that SMRs are a viable option in the mid-2020s. In addition to accomplishing the public benefit from SMR deployment, the federal government would receive a return on investment through taxes associated with investment, job creation and economic output over the lifetime of the SMR facilities that would otherwise not exist without the US government's investment."

In February 2016 TVA said it was still developing a site at Oak Ridge for a SMR and would apply for an early site permit (ESP, with no technology identified) for Clinch River in May with a view to building up to 800 MWe of capacity there. TVA has expanded discussions from B&W to include three other light-water SMR vendors. The DOE is supporting this ESP application financially from its SMR Licensing Technical Support Program, and in February 2016 DOE said it was committed to provide $36.3 million on cost-share basis to TVA.

In February 2021 TVA published a notice of intent to prepare a programmatic environmental impact statement on the potential effects of the construction, operation and decommissioning of an advanced nuclear reactor technology park at Clinch River. The park would contain one or more advanced nuclear reactors with a total electrical output of up to 800 MWe.

Another area of small reactor development is being promoted by the DOE's Advanced Research Projects Agency – Energy (ARPA-E) set up under a 2007 act. This focuses on high-potential, high-impact energy technologies that are too early for private-sector investment. ARPA-E is now beginning a new fission programme to examine microreactor technologies, below 10 MWe. This will solicit R&D project proposals for such reactors, which must have very high safety and security margins (including autonomous operations), be proliferation resistant, affordable, mobile, and modular. Targeted applications include remote sites, backup power, maritime shipping, military instillations, and space missions.

The DOE in 2015 established the Gateway for Accelerated Innovation in Nuclear (GAIN) initiative led by Idaho National Laboratory (INL) "to provide the new nuclear energy community with access to the technical, regulatory and financial support necessary to move new nuclear reactor designs toward commercialization. GAIN is based on feedback from the nuclear community and provides a single point of access to the broad range of capabilities – people, facilities, infrastructure, materials and data – across the Energy Department and its national laboratories." In January 2016 the DOE made grants of up to $40 million to X-energy for its Xe-100 pebble-bed HTR, and to Southern Company for the molten chloride fast reactor (MCFR) project being developed with TerraPower and Oak Ridge National Laboratory (ORNL).

In mid-2016 the DOE made GAIN grants of nuclear energy vouchers totalling $2 million including to Terrestrial Energy with Argonne National Laboratory, Transatomic Power with ORNL, and Oklo Inc with Argonne and INL for their respective reactor designs. A second round of GAIN voucher grants totalling $4.2 million was made in mid-2017, including to Terrestrial and Transatomic Power both with Argonne, Holtec's SMR Inventec for the SMR-160 at ORNL, Oklo Inc with Sandia and Argonne, and Elysium with INL and Argonne.

In April 2018, the DOE selected 13 projects to receive $60 million of cost-shared R&D funding for advance nuclear technologies, including the first awards under the US Industry Opportunities for Advance Nuclear Technology Development initiative.

In September 2018 the Nuclear Energy Innovation Capabilities Act and the Department of Energy Research and Innovation Act passed Congress. The first enables private and public institutions to carry out civilian research and development of advanced nuclear energy technologies. Specifically, the Act established the National Reactor Innovation Center to facilitate the siting of privately=funded advanced reactor prototypes at DOE sites through partnerships between the DOE and private industry. The second Act combines seven previously passed science bills to provide policy direction to the DOE on nuclear energy research and development.

In October 2018 the DOE announced that it was proposing to convert metallic high-assay low-enriched uranium (HALEU), with enrichment levels between 5% and 20% U-235, into fuel for research and development purposes. This would be at Idaho National Laboratory's Materials and Fuels Complex and/or the Idaho Nuclear Technology and Engineering Center, to support the

development of new reactor technologies with higher efficiencies and longer core lifetimes.

The US Nuclear Regulatory Commission (NRC) has released a draft white paper on its strategy for reviewing licensing applications for advanced non-light water reactor technologies. The NRC said it expects to finalize the draft paper by November, with submission of the first non-LWR application expected by December 2019. By mid-2019 the NRC had been formally notified by six reactor designers of their intention to seek design approval. These included three MSRs, one HTR, one FNR, and the Westinghouse eVinci heatpipe reactor. In December 2019 the Canadian Nuclear Safety Commission (CNSC) and the US NRC selected Terrestrial Energy's Integral Molten Salt Reactor (IMSR) for the first joint technical review of an advanced, non-light water nuclear reactor.

In May 2020 the DOE launched the Advanced Reactor Demonstration Program (ARDP) offering funds, initially $160 million, on a cost-share basis for the construction of two advanced reactors that could be operational within seven years. The ARDP will concentrate resources on designs that are "affordable" to build and operate. The programme would also extend to risk reduction for future demonstrations, and include support under the Advanced Reactor Concepts 2020 pathway for innovative and diverse designs with the potential to be commercial in the mid-2030s. Testing and assessing advanced technologies would be carried out at the Idaho National Laboratory's National Reactor Innovation Center (NRIC). The NRIC started up in August 2019 as part of the DOE's Gateway for Accelerated Innovation in Nuclear (GAIN) initiative, which aims to accelerate the development and commercialization of advanced nuclear technologies. In October 2020 grants of $80 million each were made to TerraPower and X-energy to build demonstration plants that can be operational within seven years.

In December 2020 the DOE announced initial $30 million funding under the ARDP for five US-based teams developing affordable reactor technologies to be deployed over 10-14 years: Kairos Power for the Hermes Reduced-Scale Test Reactor, a scaled-down version of its fluoride salt-cooled high temperature reactor (KP-FHR); Westinghouse for the eVinci microreactor; BWXT Advanced Technologies for the BWXT Advanced Nuclear Reactor (BANR); Holtec for its SMR-160; and Southern Company for its Molten Chloride Reactor Experiment, a 300 kWt reactor project to provide data to inform the design of a demonstration molten chloride fast reactor (MCFR) using TerraPower's technology.

The DOE plans to build the Microreactor Applications Research Validation and Evaluation (MARVEL) reactor, a 100 kWt microreactor at Idaho. It is designed to perform research and development on various operational features of microreactors to improve their integration with end-user applications and is described in the Research Reactors information page.

In November 2021, among other advanced reactor projects, the DOE funded the second phase of a study on the potential for small reactors in Puerto Rico, at two suggested sites.

NuScale had announced that the DOE in 2022 would fund Ukraine's State Scientific and Technical Center for Nuclear and Radiation Safety to conduct an independent review of NuScale Power's safety analysis report for its SMR technology. The review would be accessible to any Ukrainian utility interested in deploying an SMR.

In August 2022 the DOE's Nuclear Energy University Program granted funds to Core Power, the MIT Energy Initiative and Idaho National Laboratory (INL) to research the economic and environmental benefits of floating advanced nuclear power generation.

United Kingdom

The UK government in 2014 published a report on SMR concepts, feasibility and potential in the UK. It was produced by a consortium led by the National Nuclear Laboratory (NNL). Following this, a second phase of work is intended to provide the technical, financial and economic evidence base required to support a policy decision on SMRs. If a future decision was to proceed with UK development and deployment of SMRs, then further work on the policy and commercial approach to delivering them would need to be undertaken, which could lead to a technology selection process for UK generic design assessment (GDA).

In March 2016 the UK Department of Energy & Climate Change (DECC) called for expressions of interest in a competition to identify the best value SMR for the UK. This relates to a government announcement in November 2015 that it would invest at least £250 million over five years in nuclear R&D including SMRs. DECC said the objective of the initial phase was "to gauge market interest among technology developers, utilities, potential investors and funders in developing, commercializing and financing SMRs in the UK." It said the initial stage would be a "structured dialogue"

between the government and participants, using a published set of criteria, including that the SMR design must “be designed for manufacture and assembly, and … able to achieve in-factory production of modular components or systems amounting to a minimum of 40% of the total plant cost.”

In December 2017, the Department for Business, Energy & Industrial Strategy (BEIS), DECC's successor department, announced that the SMR competition had been closed. Instead, a new two-phase advanced modular reactor competition was launched, designed to incorporate a wider range of reactor types. Total funding for the Advanced Modular Reactor (AMR) Feasibility and Development (F&D) project is up to £44 million, and 20 bids had been received by the initial deadline of 7 February 2018. In September 2018 it was announced that the following eight organisations were awarded contracts up to £300,000 to produce feasibility studies for the first phase of the AMR F&D project: Advanced Reactor Concepts (ARC-100); DBD (representing China's Institute of Nuclear and New Energy Technology's HTR-PM); LeadCold (SEALER-UK); Moltex Energy (Stable Salt Reactor); Tokamak Energy (compact spherical modular fusion reactor); U-Battery Developments (U-Battery); Ultra Safe Nuclear (Micro-Modular Reactor); and Westinghouse (Westinghouse LFR).

In July 2020, under its AMR programme, BEIS awarded £10 million to each of: Westinghouse, for its 450 MWe LFR; U-Battery consortium for its 4 MWe HTR; and Tokamak Energy for its compact fusion reactor project. A further £5 million will be for British companies and start-ups to develop new ways of manufacturing advanced nuclear parts for modular reactor projects both at home and abroad. Another £5 million is to strengthen the country’s nuclear regulatory regime as it engages with advanced nuclear technologies such as these.

In March 2019 BEIS released a 2016 report on microreactors that defined them as having a capacity up to 100 MWt/30 MWe, and projecting a global market for around 570 units of an average 5 MWe by 2030, total 2850 MWe. It notes that they are generally not water-moderated or water cooled, but "use a compact reactor and heat exchange arrangement, frequently integrated in a single reactor vessel." Most are HTRs.

In 2015 Westinghouse had presented a proposal for a “shared design and development model" under which the company would contribute its SMR conceptual design and then partner with UK government and industry to complete, license and deploy it. The partnership would be structured as a UK-based enterprise jointly

owned by Westinghouse, the UK government and UK industry. In October 2016 the company said it would work with UK shipbuilder Cammell Laird as well as the UK's Nuclear Advanced Manufacturing Research Centre (NAMRC) on a study to explore potential design efficiencies to reduce the lead times of its SMR.

NuScale has said that it aims to deploy its SMR technology in the UK with UK partners, so that the first of its units could be in operation by the mid-2020s. In September 2017 the company released its five-point UK SMR action plan. Rolls-Royce submitted a detailed design to the government for a 220 MWe SMR unit.

In November 2021 the UK government announced that it would contribute £210 million in grant funding to Rolls-Royce SMR to match private investment in this venture. Rolls-Royce Group, BNF Resources UK and Exelon Generation will invest £195 million over about three years in it. Rolls-Royce said the SMR business, which will continue to seek further investment, will now "proceed rapidly with a range of parallel delivery activities, including entry to the UK generic design assessment (GDA) process and identifying sites for the factories which will manufacture the modules that enable onsite assembly of the power plants." The reactor is designed for hydrogen and synthetic fuel manufacturing as well as electricity generation. The Rolls-Royce SMR consortium, involving many of the major UK engineering firms, aims to build 16 reactors, each a pressurized water type of 470 MWe.

In November 2022, Rolls-Royce announced that it had identified four priority locations to build SMR-based power stations in the UK, including Trawsfynydd, Wylfa, and Oldbury. The locations are all on land owned by the UK Nuclear Decommissioning Authority (NDA). Before the NDA commits to the SMR development, approval must first be granted by the Department of Business, Energy, and Industrial Strategy.

Canada

A June 2016 report for the Ontario Ministry of Energy focused on nine designs under 25 MWe for off-grid remote sites. All had a medium level of technology readiness and were expected to be competitive against diesel. Two designs were integral PWRs of 6.4 and 9 MWe, three were HTRs of 5, 8 and 16 MWe, two were sodium-cooled fast reactors (SFRs) of 1.5/2.8 and 10 MWe, one was a lead-cooled fast reactor (LFR) of 3-10 MWe, and one was an MSR of 32.5 MWe. Four were under 5 MWe (an SFR, LFR, and two

HTRs). Ontario distinguishes ‘grid scale’ SMRs above 25 MWe from these (very) small-scale reactors.

The Canadian Nuclear Safety Commission (CNSC) has been conducting pre-licensing vendor design reviews – an optional service to assess a nuclear power plant design based on a vendor's reactor technology – for a number of small reactors with capacities in the range of 3-300 MWe.

* Terrestrial Energy’s IMSR; USNC’s MMR-5 and MMR-10; ARC Nuclear’s ARC-100; Moltex’s Stable Salt Reactor; SMR’s SMR-160; U-Battery's U-Battery, GE Hitachi's BWRX-300; X-energy's Xe-100; Westinghouse's eVinci micro.

In June 2017 Canadian Nuclear Laboratories (CNL) invited expressions of interest in SMRs. This resulted in many responses, including 19 for siting a demonstration or prototype reactor at a CNL-managed site. CNL aims to have a new SMR at its Chalk River site by 2026. Global First Power with its partners Ontario Power Generation and Ultra-Safe Nuclear Corporation was the first to get to the third stage of CNL’s siting evaluation, with its MMR, a 5 MWe HTR. In February 2019 CNL announced that StarCore Nuclear and Terrestrial Energy had qualified to enter the due diligence (second) stage of its siting evaluation for their 14 MWe HTR and 195 MWe IMSR respectively.

In November 2019 CNL announced that Kairos Power, Moltex Canada, Terrestrial Energy and Ultra Safe Nuclear Corporation (USNC) had been selected as the first recipients of support under its Canadian Nuclear Research Initiative (CNRI). This is designed to accelerate SMR deployment by enabling research and development on particular projects and connecting global vendors of SMR technology with the facilities and expertise within Canada's national nuclear laboratories. Recipients are expected to match the value contributed by CNL either in monetary or in-kind contributions.

In November 2018 the Canadian government released its SMR Roadmap, a 10-month nationwide study of SMR technology. The report concludes that Generation IV SMR development is a response to market forces for "smaller, simpler and cheaper" nuclear energy, and the large global market for this technology will be "driven not just by climate change and clean energy policies, but also by the imperatives of energy security and access." In October 2020 the Minister for Innovation, Science & Industry announced a C$20 million investment in Terrestrial Energy to accelerate

development of its Integral Molten Salt Reactor (IMSR), the first grant from Canada's Strategic Innovation Fund.

In December 2019 Saskatchewan and New Brunswick agreed to work with Ontario in promoting SMRs to "unlock economic potential across Canada, including rural and remote regions" in line with the national SMR Roadmap. In August 2020 Alberta joined in, flagging the potential for SMRs to be used for the province's northern oil sands industry. The agreement is to also address key issues for SMR deployment including technological readiness, regulatory frameworks, economics and financing, nuclear waste management and public and indigenous engagement. In 2021 Alberta's largest oil sands producers formed an alliance to consider ways to achieve net zero greenhouse gas emissions by 2050, with SMRs being part of the means.

In October 2020 Ontario Power Generation (OPG) announced that it would take forward engineering and design work with three developers of grid-scale SMRs – GE Hitachi (GEH), Terrestrial Energy and X-energy – to support remote area energy needs. The focus is on GEH's 300 MWe BWRX-300, Terrestrial's 192 MWe Integral Molten Salt Reactor, and X-energy's 80 MWe Xe-100 high-temperature SMRs. All three are in phase 2 of the CNSC's vendor design review process. GEH is setting up a Canadian supply chain for its BWRX-300.

In November 2020 New Brunswick Power and Moltex Energy were joined by ARC Canada in setting up an SMR vendor cluster at Point Lepreau, and in March 2021 the Canadian government announced C$56 million support for this, mostly for the Moltex Stable Salt Reactor – Wasteburner (SSR-W) project.

China

The most advanced small modular reactor project is in China, where Chinergy is starting to build the 210 MWe HTR-PM, which consists of twin 250 MWt high-temperature gas-cooled reactors (HTRs) which build on the experience of several innovative reactors in the 1960s to 1980s.

CNNC New Energy Corporation, a joint venture of CNNC (51%) and China Guodian Corp, is promoting the ACP100 reactor. A preliminary safety analysis report for a single unit demonstration plant at Changjiang was approved in April 2020.

However, China is also developing small district heating reactors of 100 to 200 MWt capacity which may have a strong potential evaluated at around 400 units. The heat market is very large in northern China, now almost exclusively served by coal, causing serious pollution, particularly by dust, particulates, sulfur, and nitrogen oxides.

Overall SMR research and development in China is very active, with vigorous competition among companies encouraging innovation.

Other

Urenco has called for European development of very small – 4 MWe – 'plug and play' inherently-safe reactors based on graphite-moderated HTR concepts. It is seeking government support for a prototype "U-Battery" which would run for 5-10 years before requiring refuelling or servicing.

Already operating in a remote corner of Siberia are four small units at the Bilibino co-generation plant. These four 62 MWt (thermal) units are an unusual graphite-moderated boiling water design with water/steam channels through the moderator. They produce steam for district heating and 11 MWe (net) electricity each, remote from any grid. They are the world's smallest commercial power reactors and have performed well since 1976, much more cheaply than fossil fuel alternatives in the severe climate of this Arctic region, but are due to be retired by 2023.

Looking ahead, and apart from its barge-mounted ones, Rosatom is not positive about small reactors generally.

Also in the small reactor category are the Indian 220 MWe pressurized heavy water reactors (PHWRs) based on Canadian technology, and the Chinese 300-325 MWe PWR such as built at Qinshan Phase I and at Chashma in Pakistan, and now called CNP-300. The Nuclear Power Corporation of India (NPCIL) is now focusing on 540 MWe and 700 MWe versions of its PHWR, and is offering both 220 and 540 MWe versions internationally. These small established designs are relevant to situations requiring small to medium units, though they are not state of the art technology.

Another significant line of development is in very small fast reactors of under 50 MWe. Some are conceived for areas away from transmission grids and with small loads; others are designed to operate in clusters in competition with large units.

Other, mostly larger new designs are described in the information page on Advanced Nuclear Power Reactors.

In December 2019 CEZ in the Czech Republic said it was focusing on 11 SMR designs including these seven: Rosatom's RITM-200, GE Hitachi Nuclear Energy's BWRX-300, NuScale Power's SMR, China National Nuclear Corporation's ACP100, Argentina's CAREM, the South Korean SMART, and Holtec International's SMR-160.

Notes & references

General sources

Small Modular Reactors Catalogue 2024, International Atomic Energy Agency (October 2024)
The NEA Small Modular Reactor Dashboard: Third Edition (September 2025)

Appendix

Military developments of small power reactors from 1950s

US experience and plans

About five decades ago the US Army built eight reactors, five of them portable or mobile. PM1 successfully powered a remote air/missile defence radar station on a mountain top near Sundance, Wyoming for six years to 1968, providing 1 MWe. At Camp Century in northern Greenland the 10 MWt, 1.56 MWe plus 1.05 GJ/hr PM-2A was assembled from prefabricated components, and ran from 1960-64 on high-enriched uranium fuel. Another was the 9 MWt, 1.5 MWe (net) PM-3A reactor which operated at McMurdo Sound in Antarctica from 1962-72, generating a total of 78 million kWh and providing heat. It used high-enriched uranium fuel and was refuelled once, in 1970. MH-1A was the first floating nuclear power plant operating in the Panama Canal Zone from 1968-77 on a converted Liberty ship. It had a 45 MWt/10 MWe (net) single-loop PWR which used low-enriched uranium (4-7%). It used 541 kg of U-235 over ten years and provided power for nine years at 54% capacity factor.

ML-1 was a smaller and more innovative 0.3 MWe mobile power plant with a water-moderated HTR using pressurized nitrogen at 650°C to drive a Brayton closed cycle gas turbine. It used HEU in a cluster of 19 pins, the core being 56 cm high and 56 cm diameter. It was tested over 1962-66 in Idaho. It was about the size of a standard shipping container and was truck-mobile and air-transportable, with 12-hour set-up. The control unit was separate, to be located 150 m away.

All these were outcomes of the Army Nuclear Power Program (ANPP) for small reactor development – 0.1 to 40 MWe – which ran from 1954-77. ANPP became the Army Reactor Office (ARO) in 1992. More recently (2010) the DEER (Deployable Electric Energy Reactor) was being commercialized by Radix Power & Energy for forward military bases or remote mining sites.

A 2018 report from the US Army analysed the potential benefits and challenges of mobile nuclear power plants (MNPPs) with very small modular reactor (vSMR) technology. This followed a 2016 report on Energy Systems for Forward/Remote Operating Bases. The purpose is to reduce supply vulnerabilities and operating costs while providing a sustainable option for reducing petroleum demand and consequent vulnerability. MNPPs would be portable by truck or large aircraft and if abroad, returned to the USA for refuelling after 10-20 years. They would load-follow and run on low-enriched uranium (<20%), probably as TRISO (tristructural-isotropic) fuel in high-temperature gas-cooled reactors (HTRs).

In January 2019 the Department of Defense (DOD) Strategic Capabilities Office solicited proposals for a 'small mobile reactor' design which could address electrical power needs in rapid response scenarios – Project Pele. These would make domestic infrastructure resilient to an electrical grid attack and change the logistics of forward operating bases, both by making more energy available and by simplifying fuel logistics needed to support existing, mostly diesel-powered, generators. They would also enable a more rapid response during humanitarian assistance and disaster relief operations. "Small mobile nuclear reactors have the potential to be an across-the-board strategic game changer for the DOD by saving lives, saving money, and giving soldiers in the field a prime power source with increased flexibility and functionality." The reactors need to be designed to be operated by a crew of six, with one fully qualified engineer and a single operator on duty at all times.

Each reactor should be an HTR with high-assay low-enriched uranium (HALEU) TRISO fuel and produce a threshold power of 1-

10 MWe for at least three years without refuelling. It must weigh less than 40 tonnes and be sized for transportability by truck, ship, and C-17 aircraft. Designs must be "inherently safe", ensuring that a meltdown is "physically impossible" in various complete failure scenarios such as loss of power or cooling, and must use ambient air as their ultimate heat sink, as well as being capable of passive cooling. The reactor must be capable of being installed to the point of "adding heat" within 72 hours and of completing a planned shutdown, cool down, disconnect and removal of transport in under seven days. The DOD announced its preparation of an environmental impact statement for the reactor in March 2020, and awarded $12-14 million contracts to three companies for initial design work. Then BWXT Advanced Technologies and X-energy were selected in March 2021 to develop a final engineering design by March 2022. Westinghouse has dropped out, and one of the two companies may be commissioned in 2022 to build a prototype reactor.

The DOD in March 2021 said Project Pele is on track for full power testing of a mobile reactor in 2023, with outdoor mobile testing of a prototype microreactor built at Idaho National Laboratory or Oak Ridge National Laboratory in 2024. The programme is also intended to spur commercial development of HTRs. In September 2021 the DOD issued a draft environmental impact statement for the construction and demonstration operation of a prototype mobile microreactor.

In October the US Air Force announced that its first microreactor would be at Eielson air force base in Alaska, near Fairbanks, to be operational in 2027. This does not appear to be part of Project Pele. The base has its own 15 MWe coal-fired power station already, with a railway to supply it with fuel.

Russian experience

The Joint Institute for Power Engineering and Nuclear Research (Sosny) in Belarus built two Pamir-630D truck-mounted small air-cooled nuclear reactors in 1976, during the Soviet era. The entire plant required several trucks. This was a 5 MWt/0.6 MWe HTR reactor using 45% enriched fuel with zirconium hydride moderator and driving a gas turbine with dinitrogen tetroxide through the Brayton cycle. After some operational experience the Pamir project was scrapped in 1985-86. It had been preceded by the 1.5 MWe TES-3, a PWR mounted on four heavy tank chassis, each self-propelled, with the modules (reactor, steam generator, turbine, control) coupled onsite. The prototype started up in 1961 at Obninsk, operated to 1965, and was abandoned in 1969.

Since 2010 Sosny has been involved with Luch Scientific Production Association (SRI SIA Luch) and Russia's N.A. Dollezhal Research and Development Institute of Power Engineering (NIKIET or RDIPE) to design a small transportable nuclear reactor. The new design will be an HTR concept similar to Pamir but about 2.5 MWe.

A small Russian HTR which was being developed by NIKIET is the Modular Transportable Small Power Nuclear Reactor (MTSPNR) for heat and electricity supply of remote regions. It is described as a single circuit air-cooled HTR with closed cycle gas turbine. It uses 20% enriched fuel and is designed to run for 25 years without refuelling. A twin unit plant delivers 2 MWe and/or 8 GJ/h. It is also known as GREM. No recent information is available, but an antecedent is the Pamir, from Belarus. More recently NIKIET has described the ATGOR – a transportable HTR with up to six parallel commercial gas-turbine engines with two independent heat sources (a nuclear reactor and a start-up diesel fuelled combustor).

Another NIKIET project is the 6 MWt, 1 MWe Vityaz modular integral light water reactor with two turbine generators, which is transportable as four modules of up to 60 tonnes.

In 2015 it was reported that the Russian defence ministry had commissioned the development of small mobile nuclear power plants for military installations in the Arctic. A pilot project being undertaken by Innovation Projects Engineering Company (IPEC) is a mobile low-power nuclear unit to be mounted on a large truck, tracked vehicle or a sledged platform. Production models will need to be capable of being transported by military cargo jets and heavy cargo helicopters, such as the Mil Mi-26. They need to be fully autonomous and designed for years-long operation without refuelling, with a small number of personnel, and remote control centre. It is assumed but not confirmed that these reactors will be the MTSPNR.